


# DATA VISUALISATION

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# Outline

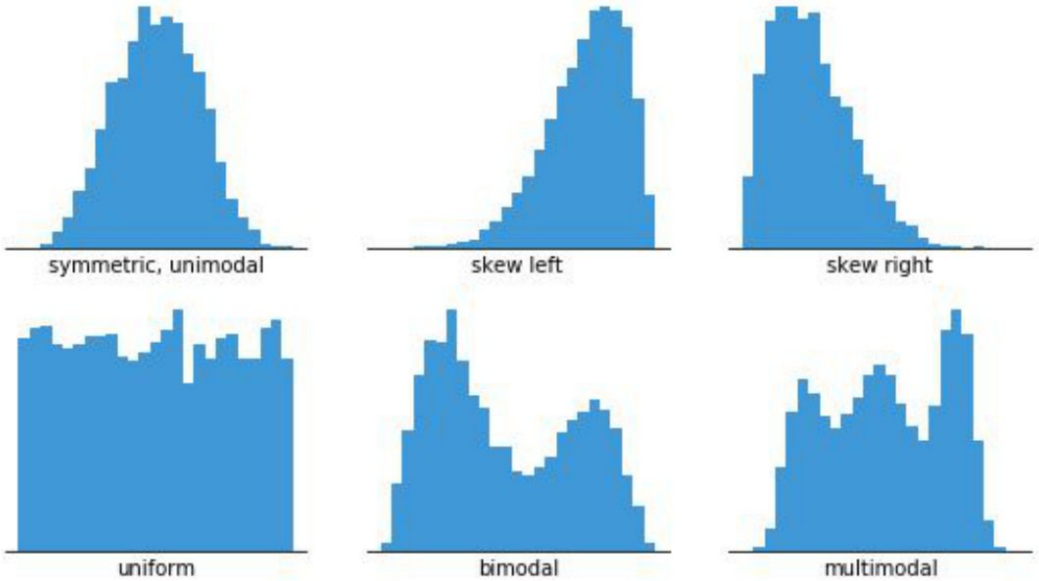
- Basic plots
- Histogram
- Bar plot
- Box plot
- Scatter plot
- Grammar of graphics
- Basic template
- Mini exercise

# Basic plots

- Histogram
- Bar plot
- Box plot
- Scatter plot

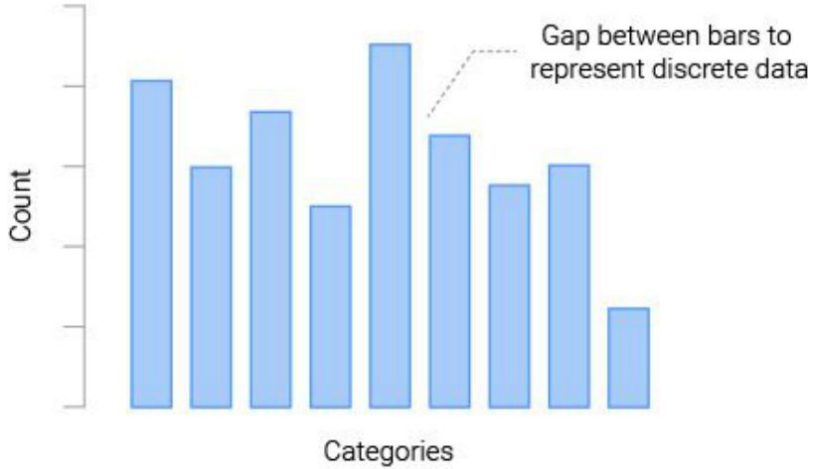
# Histogram

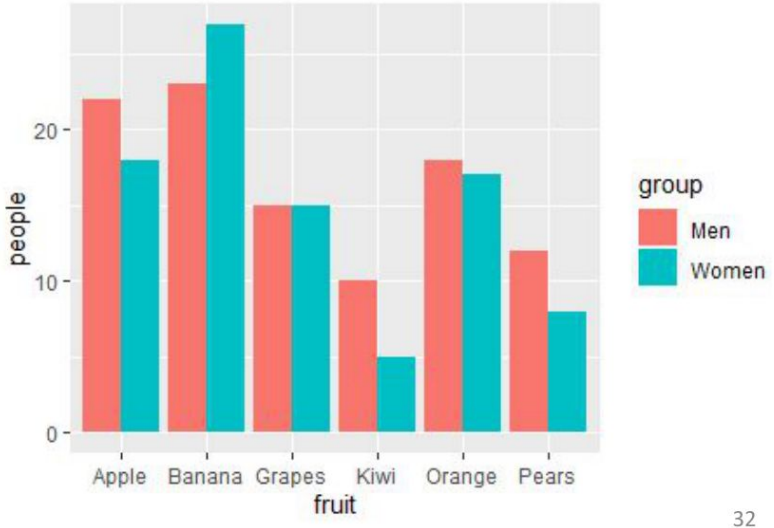
- Describes the distribution of numeric values



# Bar plot

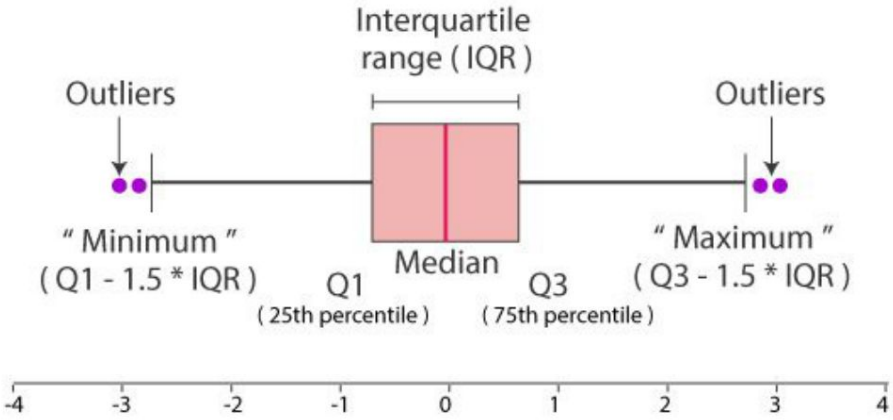
- Describes the distribution of a categorical variable



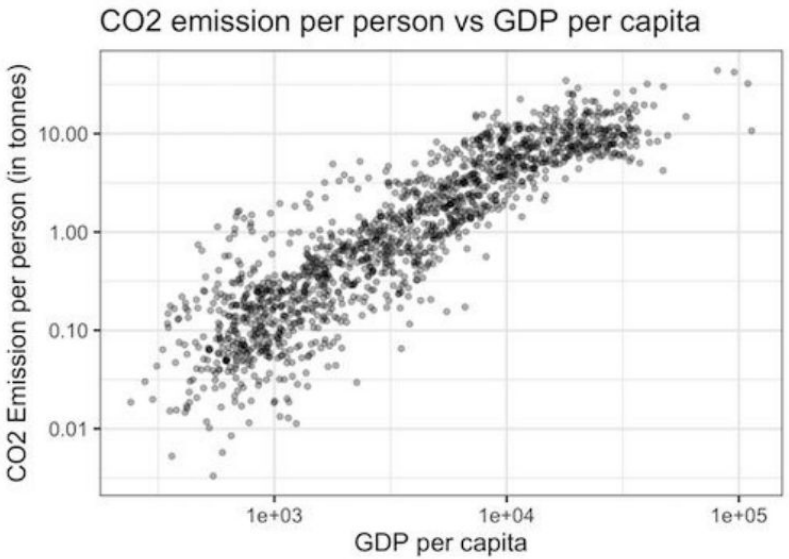
# Box plot

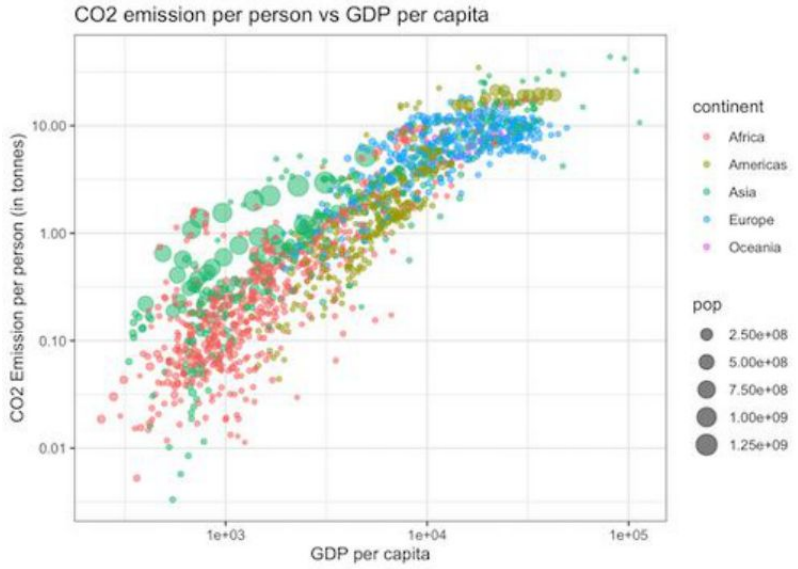
- Also known as box and whisker plot
- Can be used to detect outliers: values that deviate too much from other values in the dataset



# Scatter plot

- Represents values for two numeric variables



# Grammar of graphics

- ggplot2 builds a plot from layers
- Data: the data frame
- Aesthetics: map variables to x, y, colour
- Geometries: points, bars, boxes, lines
- Layers are added with +

# Basic template

- Every plot follows one pattern
- Start with ggplot(), add layers with +

```
ggplot(data, aes(x = var1, y = var2)) +
  geom_point()
```

# Choosing the right plot

- One numeric: histogram
- One categorical: bar plot
- Numeric by category: box plot
- Two numeric: scatter plot
- Two categorical: stacked / dodged bar

# Mini exercise (15 min)

- Use mtcars dataset (pre-existing dataset in R)
  a. Histogram of one numerical variable
  b. Box plot mpg by am
  c. Scatter plot 2 numerical variables with geom_smooth()
     - Add labs() and theme_minimal()
     - Save the plot in jpg



# Any question?